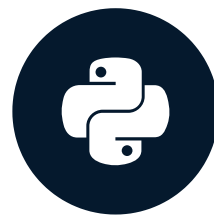


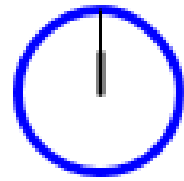
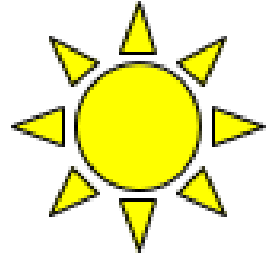
UTC offsets

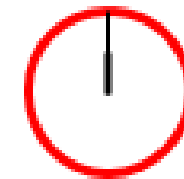
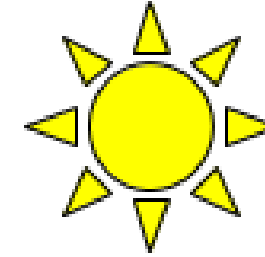
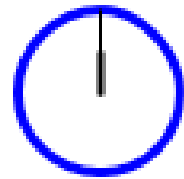
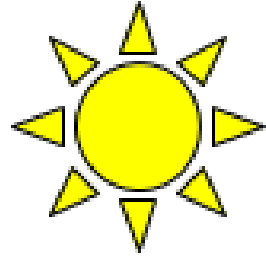
WORKING WITH DATES AND TIMES IN PYTHON

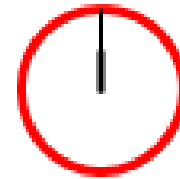
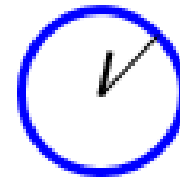
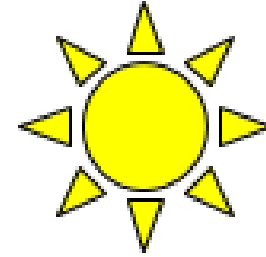
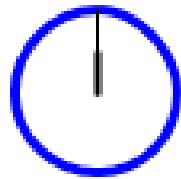
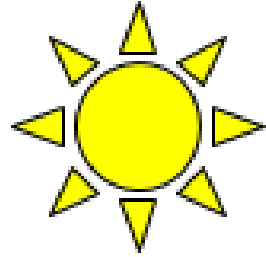


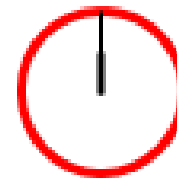
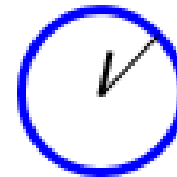
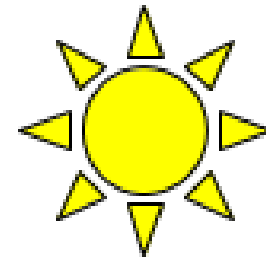
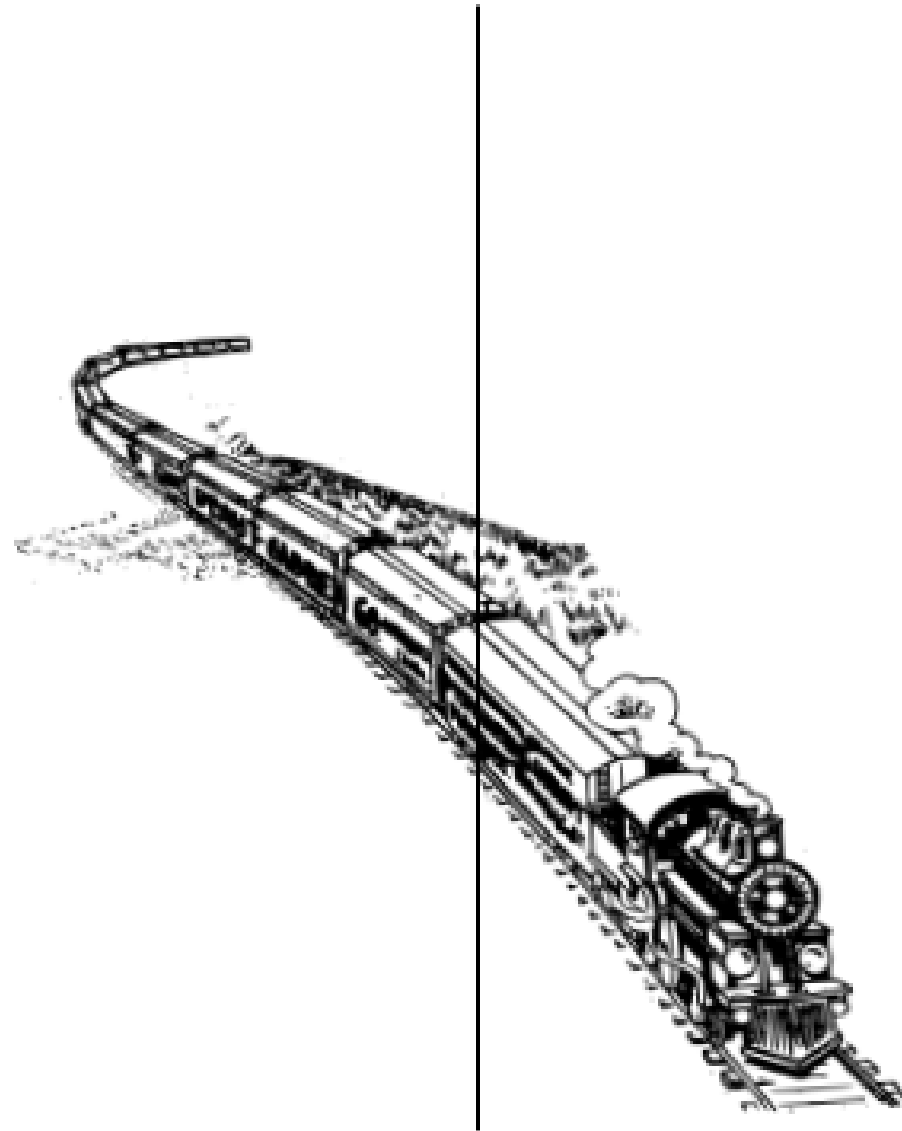
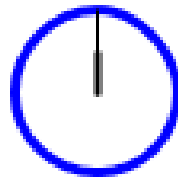
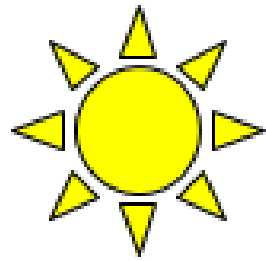
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Data Scientist and Author

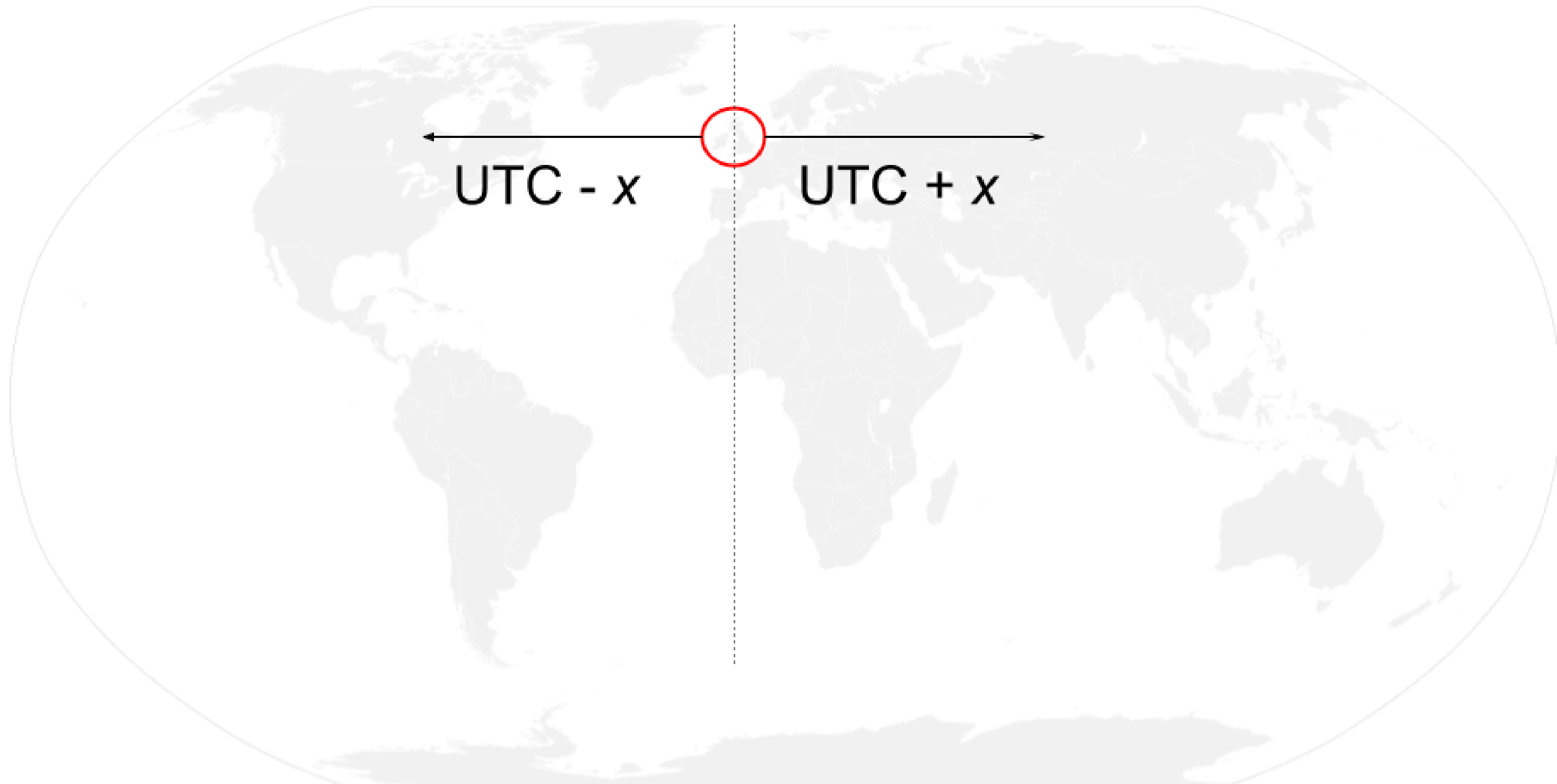












UTC

```
# Import relevant classes  
from datetime import datetime, timedelta, timezone
```

UTC

```
# Import relevant classes
from datetime import datetime, timedelta, timezone

# US Eastern Standard time zone
ET = timezone(timedelta(hours=-5))
# Timezone-aware datetime
dt = datetime(2017, 12, 30, 15, 9, 3, tzinfo = ET)

print(dt)
```

```
2017-12-30 15:09:03-05:00
```

UTC

```
# India Standard time zone  
IST = timezone(timedelta(hours=5, minutes=30))  
# Convert to IST  
print(dt.astimezone(IST))
```

```
2017-12-31 01:39:03+05:30
```

Adjusting timezone vs changing tzinfo

```
print(dt)
```

```
2017-12-30 15:09:03-05:00
```

```
print(dt.replace(tzinfo=timezone.utc))
```

```
2017-12-30 15:09:03+00:00
```

```
# Change original to match UTC
```

```
print(dt.astimezone(timezone.utc))
```

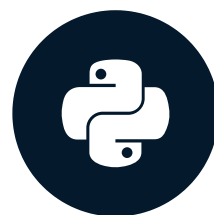
```
2017-12-30 20:09:03+00:00
```

UTC Offsets

WORKING WITH DATES AND TIMES IN PYTHON

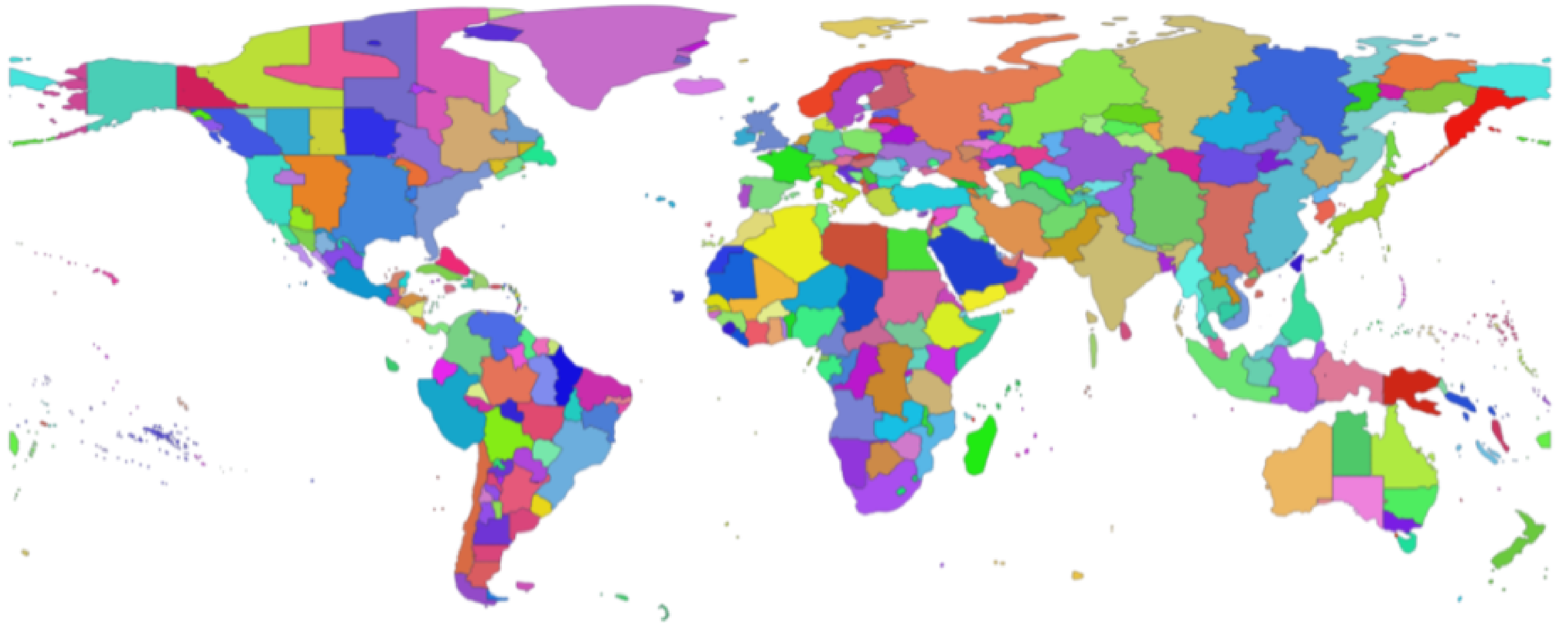
Time zone database

WORKING WITH DATES AND TIMES IN PYTHON



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Time zone database

```
# Imports  
from datetime import datetime  
from dateutil import tz
```

tz database

Time zone database

```
# Imports
from datetime import datetime
from dateutil import tz

# Eastern time
et = tz.gettz('America/New_York')
```

tz database

- Format: 'Continent/City'

Time zone database

```
# Imports
from datetime import datetime
from dateutil import tz

# Eastern time
et = tz.gettz('America/New_York')
```

tz database

- Format: 'Continent/City'
- Examples:
 - 'America/New_York'
 - 'America/Mexico_City'
 - 'Europe/London'
 - 'Africa/Accra'

Time zone database

```
# Last ride  
last = datetime(2017, 12, 30, 15, 9, 3, tzinfo=et)
```

```
print(last)
```

```
2017-12-30 15:09:03-05:00
```

Time zone database

```
# Last ride
last = datetime(2017, 12, 30, 15, 9, 3, tzinfo=et)
print(last)
```

```
2017-12-30 15:09:03-05:00
```

```
# First ride
first = datetime(2017, 10, 1, 15, 23, 25, tzinfo=et)
print(first)
```

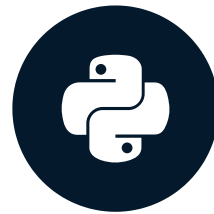
```
2017-10-01 15:23:25-04:00
```

Time zone database

WORKING WITH DATES AND TIMES IN PYTHON

Starting Daylight Saving Time

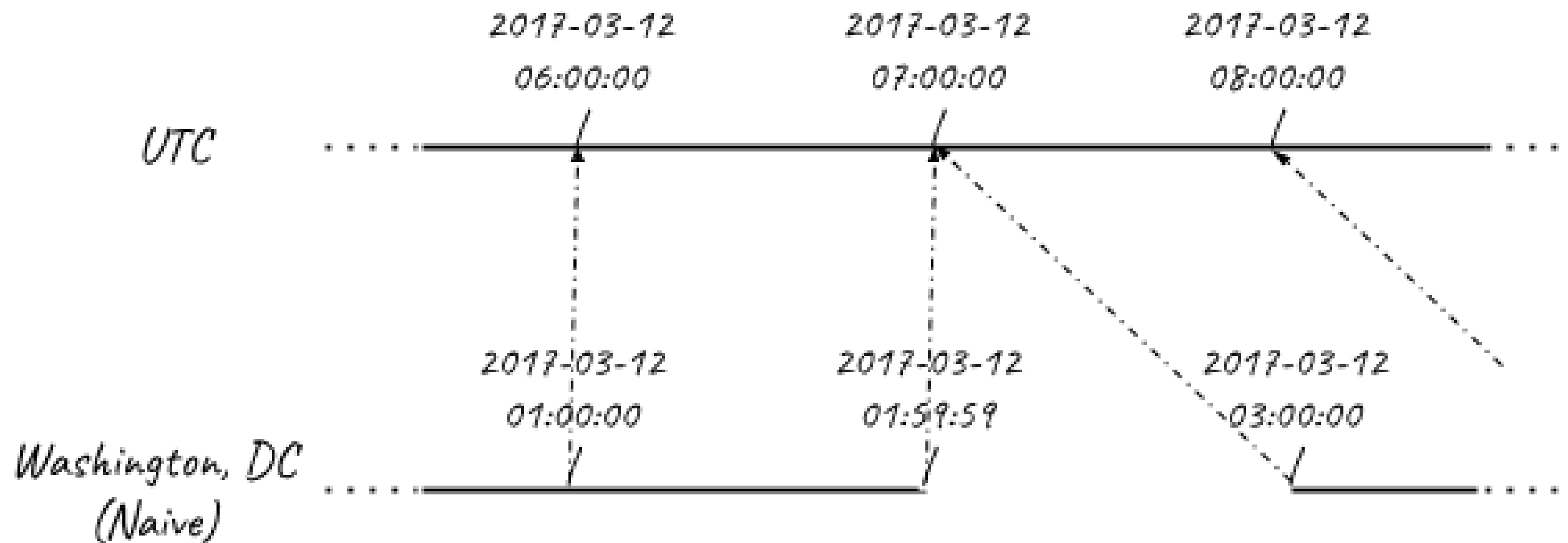
WORKING WITH DATES AND TIMES IN PYTHON



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Washington, DC (Naive) 2017-03-12 01:00:00 2017-03-12 01:59:59 2017-03-12 03:00:00



Start of Daylight Saving Time

```
spring_ahead_159am = datetime(2017, 3, 12, 1, 59, 59)
spring_ahead_159am.isoformat()
```

```
'2017-03-12T01:59:59'
```

```
spring_ahead_3am = datetime(2017, 3, 12, 3, 0, 0)
spring_ahead_3am.isoformat()
```

```
'2017-03-12T03:00:00'
```

```
(spring_ahead_3am - spring_ahead_159am).total_seconds()
```

```
3601.0
```

Start of Daylight Saving Time

```
from datetime import timezone, timedelta
```

```
EST = timezone(timedelta(hours=-5))
```

```
EDT = timezone(timedelta(hours=-4))
```

Start of Daylight Saving Time

```
spring_ahead_159am = spring_ahead_159am.replace(tzinfo = EST)  
spring_ahead_159am.isoformat()
```

```
'2017-03-12T01:59:59-05:00'
```

```
spring_ahead_3am = spring_ahead_3am.replace(tzinfo = EDT)  
spring_ahead_3am.isoformat()
```

```
'2017-03-12T03:00:00-04:00'
```

```
(spring_ahead_3am - spring_ahead_159am).seconds
```

```
1
```

Start of Daylight Saving Time

Using `dateutil`

```
# Import tz
from dateutil import tz

# Create eastern timezone
eastern = tz.gettz('America/New_York')

# 2017-03-12 01:59:59 in Eastern Time (EST)
spring_ahead_159am = datetime(2017, 3, 12, 1, 59, 59,
                              tzinfo = eastern)

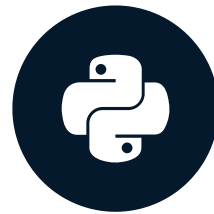
# 2017-03-12 03:00:00 in Eastern Time (EDT)
spring_ahead_3am = datetime(2017, 3, 12, 3, 0, 0,
                             tzinfo = eastern)
```

Daylight Saving

WORKING WITH DATES AND TIMES IN PYTHON

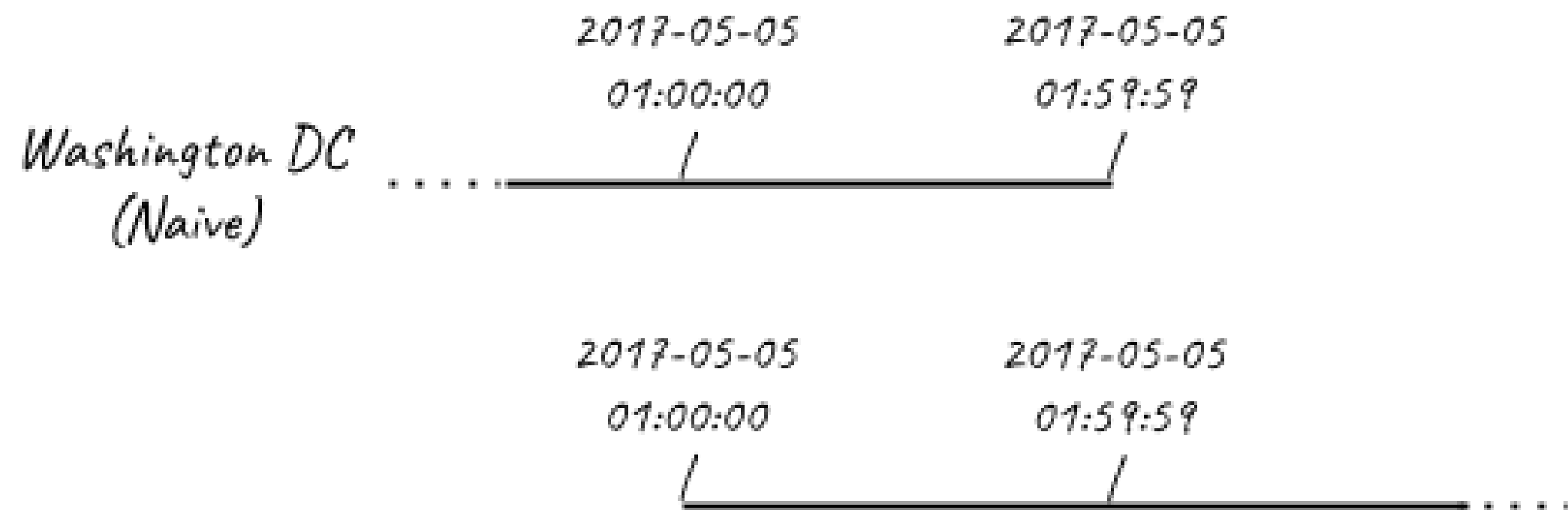
Ending Daylight Saving Time

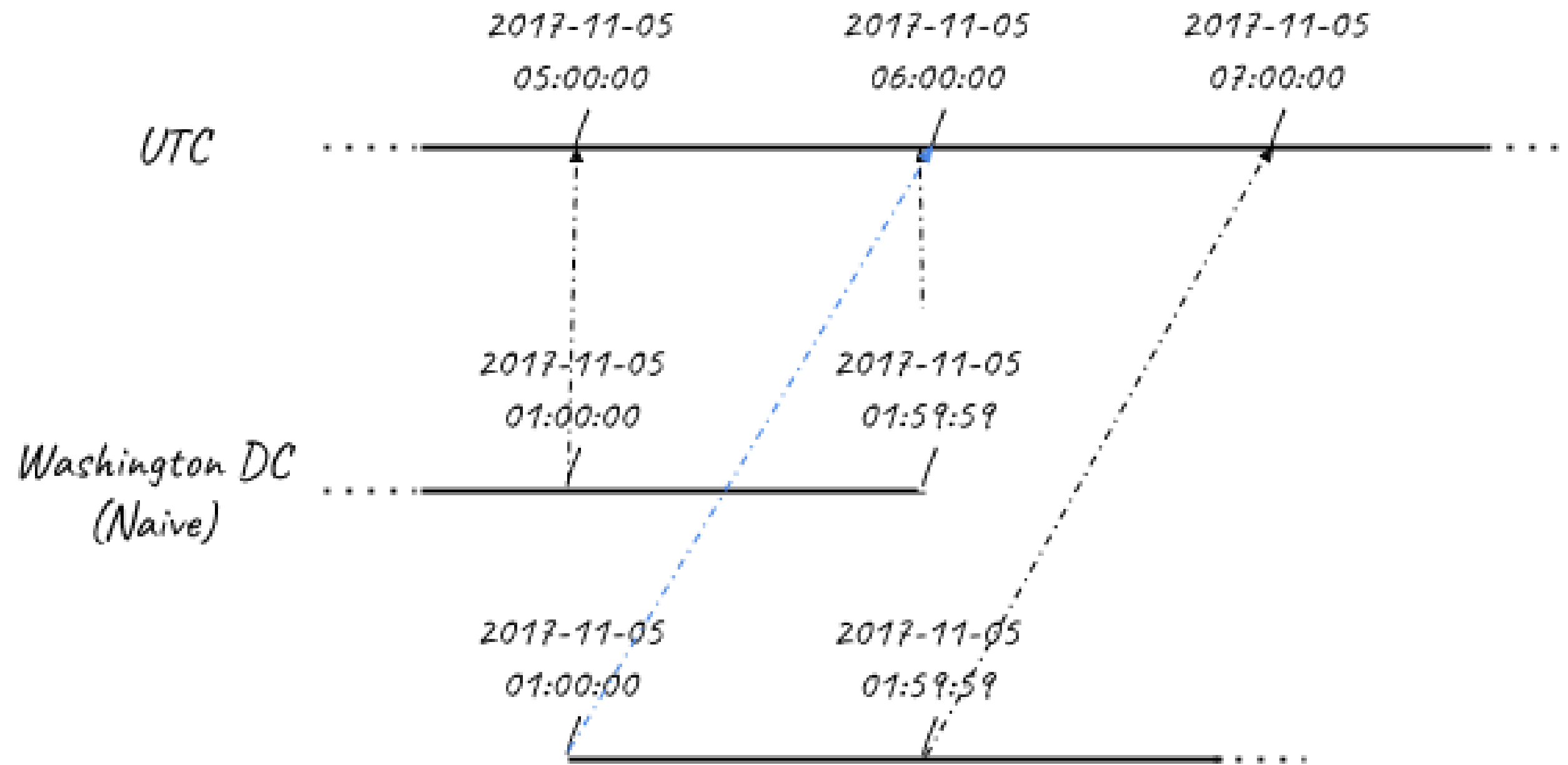
WORKING WITH DATES AND TIMES IN PYTHON



Max Shron

Data Scientist and Author





Ending Daylight Saving Time

```
eastern = tz.gettz('US/Eastern')
# 2017-11-05 01:00:00
first_1am = datetime(2017, 11, 5, 1, 0, 0,
                     tzinfo = eastern)
tz.datetime_ambiguous(first_1am)
```

True

```
# 2017-11-05 01:00:00 again
second_1am = datetime(2017, 11, 5, 1, 0, 0,
                      tzinfo = eastern)
second_1am = tz.fold(second_1am)
```

Ending Daylight Saving Time

```
(first_1am - second_1am).total_seconds()
```

```
0.0
```

```
first_1am = first_1am.astimezone(tz.UTC)  
second_1am = second_1am.astimezone(tz.UTC)  
(second_1am - first_1am).total_seconds()
```

```
3600.0
```

Ending Daylight Saving Time

WORKING WITH DATES AND TIMES IN PYTHON